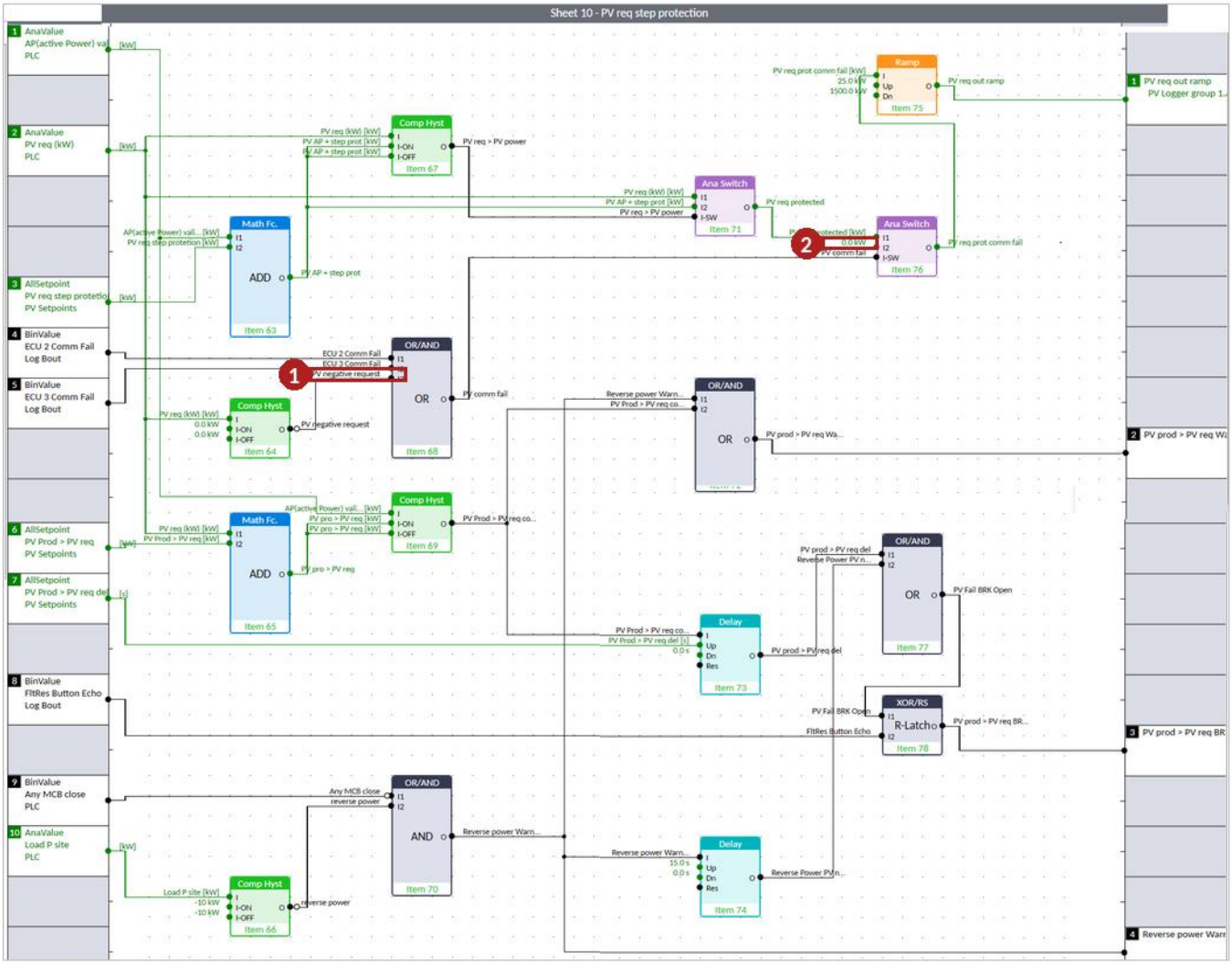
Sheet 10 as it is today - the problem

Erbil plant - InteliNeo 5500 master - PLC sheet 10 "PV req step protection"

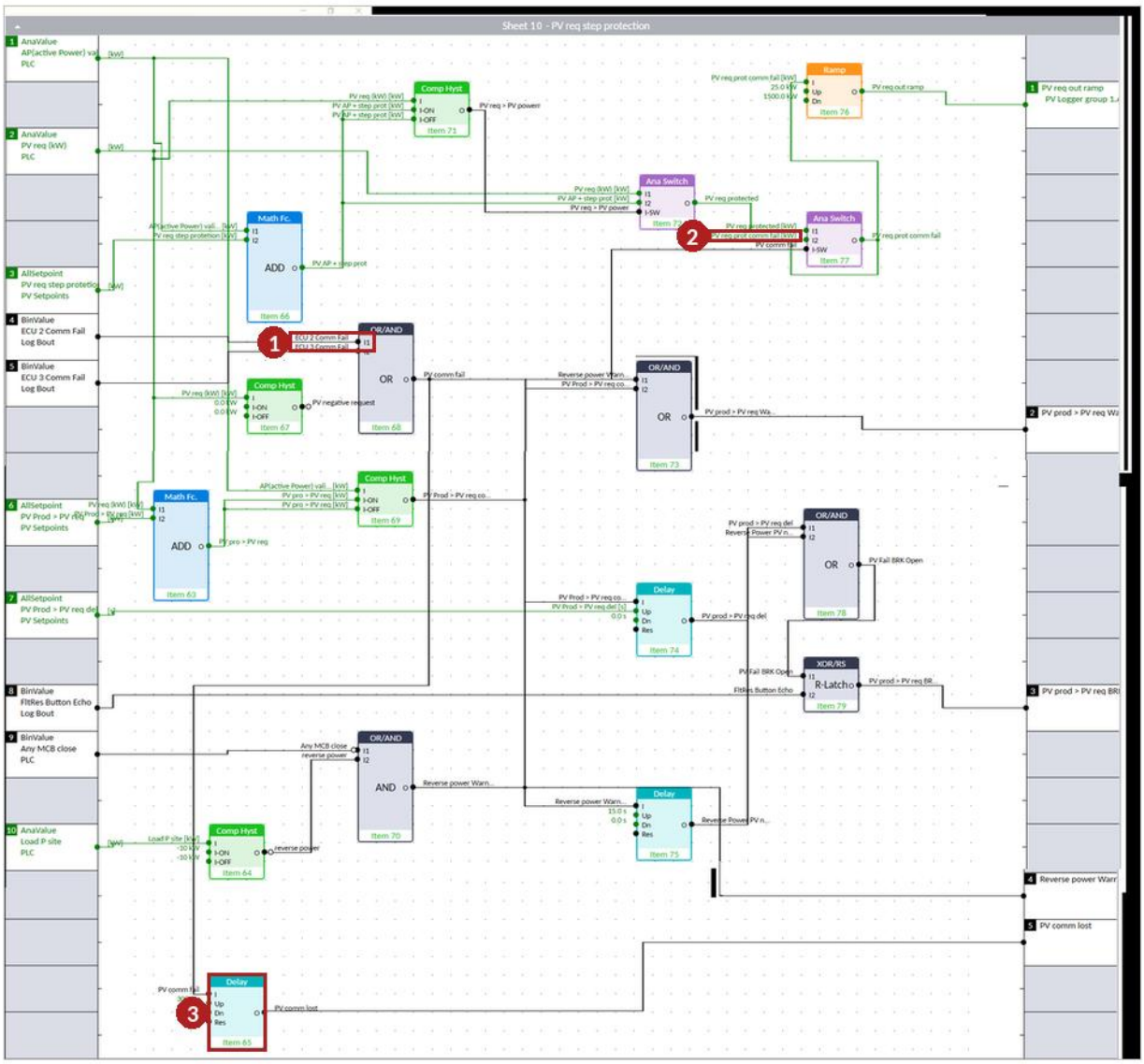


Item 76 is a switch. In its normal position the solar request passes through. When either PV logger session misses a single reply, Item 68 (1) flips it to its second position, which is a fixed 0.0 kW (2). That zero is written to the SAU as the solar setpoint.

The inverters obey within seconds. The gensets, idling at 25 %, suddenly carry the whole plant; the UPS logic sees the jump and sheds the UPS breakers; and the solar then needs about five minutes to climb back, because the request may never be more than 100 kW above what the inverters produce. The history shows this five to thirty times a day. The zero never even reaches the logger while the link is down - it arrives when the link is back, when there is nothing left to protect.

The change I request

Erbil plant - InteliNeo 5500 master - PLC sheet 10 "PV req step protection"



Three edits on this sheet, nothing else. (1) Item 68 reacts only to the two comm fail signals, ECU 2 and ECU 3; the "request below zero" input is removed, so a request of zero still goes out as zero. (2) The same switch, now Item 77, has its own output on its second input: when a reply is missed it simply repeats the last setpoint it sent, and nothing drops. (3) One new timer, Item 65, raises the flag "PV comm lost" after five minutes without the logger, for an alarm; nothing acts on it yet.

Why it should go in

With a healthy link the program behaves exactly as today - checked block by block against the plant's own history. The risk during a lost link is unchanged: the inverters keep their last setpoint in both versions, the gensets hold 25 % of their nominal in hand, the controller's own protections do not depend on the logger, and after five minutes the flag is raised. It is undone in a minute with the 14 October archive. On the recorded day of 14 October it removes three of the eleven UPS sheddings and gives back about 350 kWh of solar - and it is the precondition for running fewer gensets, where the fuel is.